

Looming as free energy: the ways to read it, and what each predicts

2026-09-22, on Jonas's instruction: "Continue pursuing the possibility of using active inference for CZBs. It may be that a looming threshold fits the data better, but if we can explain it in terms of fundamental free energy it is more anchored in fundamental science, but we should not just assume it works: we have to probe the different ways to think about it." A design note, not a result; the cards it proposes are pre-stated and run separately, and section 6 records their outcomes as they arrive. Opinions are marked.

1 What is on the table

The measurement model that fits the second cut-in study is

$$\text{share who intervene} = \text{lapse} + (1 - \text{lapse}) \times \text{GATE} \times \Phi((\log \theta - \log \theta_0) / \sigma),$$

with θ the optical expansion rate of the cut-in vehicle at the response moment (card EL.1b), the gate the probability that the vehicle's lateral clearance projected 3 s ahead falls below a minimum (card G.1), and θ_0 and σ a level and a spread. It scores 0.1027 held out; the active-inference constructions of the 09-18 to 09-22 arc scored 0.29 to 0.32. The question is not which fits better; it is whether the model that fits *is* an active-inference model written in different words, and if so, which of the several possible readings it is, because the readings make different predictions.

Card JJ.6 settled one part on 2026-09-22: **the gate is the belief**. Card JJ.1's fan, read as "the probability that the other's body is in my lane within T seconds", reproduces G.1's gate where it is closed (pre-onset 0.0354 against 0.0318, nothing fitted) and fails to reproduce its grading where it opens, because the intention posterior is binary on these cells. So the gate has an active-inference reading already: a posterior over the other's intention, propagated through the predictive model. What follows concerns the other three parts, the axis, the level and the spread, and one structural question, the horizon.

2 Four readings

Reading A, the horizon-summed expected free energy. The standard form: $G(\pi) = \sum_{\tau} E_{\tau} q[-\log C(o_{\tau})] +$ epistemic terms, over a planning horizon, with C a prior preference over observations. Cards JJ.2, JJ.2c, RE.4b, S1.6, JJ.5 and JJ.5b all read the data this way, with the preference being the released collision and braking-margin terms, the released τ^{-1} term, or an admissibility constraint. **All are anti-ordered with the response**, and the checks of the review found why: a one-sided preference summed over the horizon of a continue policy that drives through the lead counts the number of steps before contact, which grows with TTC, while participants respond to proximity. JJ.5b removed the sum (per-step rate, maximum, first step) and the inversion went with it ($p(\text{gap}) + 0.662 \rightarrow -0.228$), but no axis appeared, because the released term is a TTC quantity. **This reading is the one the data reject, and the rejection is structural: not the constants, not the functional, the horizon sum itself.** (Opinion: it should be said plainly to JJ, because it is the difference between a planning model and a comfort-zone model.)

Reading B, the free energy of the present observation. In the continuous-time formulation action minimises the free energy of *current* sensory states, $F \approx \frac{1}{2} ((\theta - \mu)/\sigma_o)^2 + \dots$, where μ is what the generative model predicts and σ_o the sensory precision. With a prior expectation that looming stays at or below a level θ_0 (a one-sided prior: $-\log C(\theta) = \frac{1}{2} ((\theta - \theta_0)/\sigma_c)^2$ for $\theta > \theta_0$, else 0), the prediction error is $(\theta - \theta_0)$ and the reflex that cancels it is to brake. A driver responds when the error is non-zero. If the driver's belief about θ (through sensory noise) or about θ_0 (a population of levels) is log-normal, $P(\text{respond}) = \Phi((\log \theta - \log \theta_0)/\sigma)$. **This reading IS the threshold model**, with the level the mean of the prior over looming and the spread its precision (or the sensory precision; §3 says how to tell). It has no horizon, so it cannot be inverted by one, and it is gated by the belief of card JJ.6 because the prior over looming applies to a vehicle that is, or is about to be, in the ego's path. It is the reading I recommend (opinion), because nothing in it is fitted that the measurement model does not already fit, and the two fitted quantities acquire a meaning.

Reading C, the one-step expected free energy, a decision. Keep the preference of B but let the driver *compare policies* at the present step: $G(\text{continue}) = \frac{1}{2} ((\theta - \theta_0)/\sigma_c)^2$, $G(\text{brake}) =$ the effort cost $e = \frac{1}{2}(a/\sigma_a)^2$ (the released control-effort term), and choose brake when $G(\text{continue}) > G(\text{brake})$. Deterministically that is a threshold at $\theta = \theta_0 + \sigma_c \sqrt{(2e)}$: *the effort term raises the level, which is a substantive prediction (a driver with a higher tolerance for braking effort responds earlier).* With the softmax policy posterior of active inference, $P(\text{brake}) = \sigma(\gamma (G(\text{continue}) - G(\text{brake})))$, a logistic in the squared excess*, not a probit in $\log \theta$. Readings B and C therefore differ in the link

function, and that is testable on the cells (card JJ.8). If the data cannot tell them apart, the difference is philosophical; if they can, one is right.

Reading D, surprise and epistemic value. $-\log p(\theta \mid \text{model})$, the surprise of the looming observation under a generative model that predicts the other holds course, and the information gain about the other's intention. On this design both are gates, not axes: the stimulus vehicles hold constant speed, so the looming a course-holding model predicts is the looming observed (surprise is zero post-onset), and the intention posterior jumps from 0.07 to 1 in every cell (card JJ.6), so the information gain is one constant post-onset and another pre-onset. Card HS.1 tested situational surprise as the onset on 2026-09-13 and it lost to the anticipatory gate; JJ.6 is the epistemic reading done properly, and it is the gate. **Nothing more to test here on these data;** on naturalistic data, where the other's speed varies, reading D becomes an axis candidate again.

3 What the readings predict differently, and the cards that test them

question	reading B	reading C	card
the link function	probit in $\log \theta$	logistic in the squared excess $(\theta - \theta_0)^2$, with an effort offset	JJ.8 : fit both with three parameters on the fixed looming axis and G.1's frozen gate; rule: held-out difference below 0.005 is "indistinguishable on this design"
the scale of the spread	multiplicative (a spread in $\log \theta$: a prior over levels, or Weber-like sensory noise)	additive if the noise is sensory in absolute θ (the released agent's <code>sigma_phidot</code> is absolute, 1e-3 rad/s)	JJ.8 also fits the probit in θ (not log); the registered R.2 script scored raw against log scales for its other axes, never for looming
what the spread is	prior precision or sensory precision, indistinguishable on one study		JJ.4's two results bear on it: the truck's criterion is sharper (a wider vehicle gives a larger θ at the same gap, so <i>absolute</i> sensory noise predicts a smaller spread in $\log \theta$ for the truck by the width ratio, about 1.3, where JJ.4 found 1.55 with about half attributable to axis geometry per the review); and drivers are sharper in the car than at a screen (sensory precision, or the criterion's)
the effort offset	none	the level rises with the driver's effort tolerance	needs per-driver effort data; not testable here, noted for the naturalistic plan
the horizon	none	none	reading A is the one with a horizon and is rejected (§2)

Card JJ.8 is the one that costs an hour and settles something. The rest is for the record.

4 The mapping, written out

For reading B with a log-normal population of levels, $\theta_0 \sim \text{LogNormal}(m, \sigma)$, and a deterministic response when $\theta > \theta_0$: $P(\text{respond} \mid \theta) = P(\log \theta_0 < \log \theta) = \Phi((\log \theta - m)/\sigma)$. The measurement model's level is $\exp(m)$, the population median of the prior's mean over looming, and spread is σ , the population spread of that mean. Card TR.1's per-driver levels are then per-driver priors, and their consistency across scenarios (+0.647) is the claim that a driver carries one prior over looming into every scenario. The percentile deliverable is a percentile of that prior over drivers. Nothing new is fitted; the words change.

For reading C with the same population and a softmax posterior with precision γ : $P(\text{brake} \mid \theta) = \int \sigma(\gamma [\frac{1}{2}((\theta - \theta_0)/\sigma_c)^2 - e]) dP(\theta_0)$. With $\gamma \rightarrow \infty$ this is reading B's probit with the level shifted by $\sigma_c \sqrt{(2e)}$. With finite γ the curve is flatter at the top (a driver who is far past the level still hesitates with probability $\sigma(-\gamma \text{large}) \rightarrow 0$, so not flatter there; flatter at the bottom: $P(\text{brake}) = \sigma(-\gamma e) > 0$ below the level, which the lapse parameter absorbs). The distinguishing feature is the shape between: quadratic growth of the argument in $\theta - \theta_0$ against linear in $\log \theta$. On a design spanning θ from 0.004 to 1.1 rad/s (two and a half decades) the two are not close, so JJ.8 has power.

5 What would make me give up the active-inference reading (opinion)

Not a worse score: reading B fits by construction. It would be a *prediction that fails*: JJ.8 finding the logistic-in-squared-excess link clearly better than the probit (then the decision reading C wins and the reflex reading B is wrong) is fine either way, both are active inference. What would hurt is the gate: if a graded intention belief (card JJ.6b) cannot reproduce G.1's graded gate however the likelihood is made consistent with the generative model, then the gate the data want is not a belief about intention but a geometric projection, and the active-inference account of emergence is decorative. And on naturalistic data, where reading D becomes an axis candidate, a looming-surprise axis that scores no better than the looming level would say the generative model adds nothing to the preference.

6 Outcomes, as they arrive

(appended by the cards; do not edit the sections above to match)

2026-09-22, the gate (cards JJ.6 to JJ.6e, worklog entry of the same night). Section 1 said "the gate is the belief" on the strength of card JJ.6; the four cards that followed sharpen it. The intention belief supplies the gate's *closed* state (pre-onset 0.0354, nothing fitted) but not its grading: at no keeping spread does a changepoint filter over the intention open as early as participants do (JJ.6b to JJ.6d), because G.1's gate grades on how soon the body reaches mine, not on whether it intends to. And that has an exact form: G.1's gate **is** a Gaussian-rate predictor's $P(\text{clearance} < m \text{ within } T)$, with $s_l = \sigma T$ (identity to $1e-16$), and with $m = 0$ it scores $0.1028 / 0.0462$ (JJ.6e, DERIVED, the circularity through $\sigma = \text{SD_VLAT}$ declared). So §1's sentence should read: **the gate is the predictive uncertainty of the generative model about the other's lateral motion, at the anticipation horizon; the intention variable is not needed.** Reading D's "epistemic value is the gate" is therefore wrong on this design: the gate is not information gain about a latent intention but the tail of a predictive distribution. Reading B is untouched.

2026-09-22, the link and the scale (card JJ.8, out/jj8_link_and_scale.md). Additive sensory noise (a probit in θ rather than $\log \theta$) is **rejected**, $+0.023$ held out: the spread is multiplicative, a spread of levels or Weber-like noise, not the released agent's absolute σ_{phidot} . The decision reading C on a linear preference is **rejected** ($+0.021$, and its fit degenerates). C on a log-scaled preference is **indistinguishable** from the reflex reading B (-0.0002), but its fit is degenerate too (level 0.0000 rad/s, offset 21: a logistic in a quadratic of $\log \theta$ imitating the probit), so it delivers no comfort-zone level. §3's first two rows are answered: the data cannot separate B from C in shape, can separate multiplicative from additive, and only B's parameters mean anything on this design. §4's mapping stands as the reading to carry.

2026-09-23, the future and the past (card JJ.12, out/jj12_sums_future_past.md). Jonas asked whether inverse τ , with its blow-up near contact, would flip the horizon sum. In closed form and on the cells: no. The sum to contact of τ^{-1} , of its square, of θ and of θ^2 are all ordered against the participants (0 of 24 matched-TTC rows; $p(\text{gap}) +0.84$ to $+1.00$), and the sum of τ^{-1} correlates -0.861 with the share, which is card RE.2's "inversion" to three decimals. Steeper costs make the sum proportional to the closing speed, which within a row is backwards. A sum over the **past** (the Farewell accumulator) is a different object and goes the human way, 24 of 24 rows: over the shown clip 0.1332 , since the onset 0.1283 with a free step gate (pre-onset 0.0271), against the instantaneous looming rate's 0.1130 . So reading A stays rejected for the future sum, and an accumulator reading, F integrated over the past, is the one construction left that could beat the instantaneous threshold on this design (query JJ12.Q1).